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(Recognised by the Indian Society of Forensic Science)

CUET-UG 2026

SAMPLE PAPER

A National Level Socio-Forensic Voluntary Student Led Educational Firm

CHEMISTRY (306)

Q1. The van’t Hoff factor for NaCl in aqueous solution is nearly: (PYQ CUET UG 2022)

A. 1

B. 2

C. 3

D. 4

Q2. Which of the following is a colligative property? (PYQ NEET 2021)

A. Viscosity

B. Osmotic pressure

C. Surface tension

D. Refractive index

Q3. The molarity of pure water at 25°C is approximately:

A. 1 M

B. 18 M

C. 55.5 M

D. 100 M

Q4. Elevation in boiling point depends upon:

A. Nature of solute

B. Number of solute particles

C. Nature of solvent only

D. Chemical nature of solute

Q5. Which solution will have the highest osmotic pressure?

A. 0.1 M glucose

B. 0.1 M NaCl

C. 0.1 M $CaCl_2$

D. 0.1 M urea

Q6. Oxidation takes place at which electrode in an electrochemical cell? (PYQ CUET UG 2022)

A. Cathode

B. Anode

C. Salt bridge

D. Electrolyte

Q7. SI unit of electrical conductivity is:

A. ohm

B. siemens m^{-1}

C. volt

D. ampere

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Q8. Which cell converts chemical energy into electrical energy?

A. Electrolytic cell

B. Galvanic cell

C. Fuel cell

D. Concentration cell

Q9. The standard electrode potential of hydrogen electrode is taken as: (PYQ NEET 2020)

A. +1.00 V

B. −1.00 V

C. 0.00 V

D. +0.50 V

Q10. In electrolysis of molten NaCl, sodium is liberated at:

A. Anode

B. Cathode

C. Both electrodes

D. Salt bridge

Q11. Unit of rate constant for a first-order reaction is: (PYQ CUET UG 2022)

A. mol L^{-1} s^{-1}

B. s^{-1}

C. L mol^{-1} s^{-1}

D. mol^2 L^{-2} s^{-1}

Q12. Half-life of a first-order reaction depends on: (PYQ NEET 2020)

A. Initial concentration

B. Rate constant

C. Temperature only

D. Pressure

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Q13. The order of reaction is determined experimentally from: (PYQ NEET 2020)

A. Balanced equation

B. Rate law

C. Stoichiometry

D. Molecularity

Q14. For a zero-order reaction, rate is independent of:

A. Time

B. Temperature

C. Concentration

D. Catalyst

Q15. Increase in temperature increases reaction rate because:

A. Activation energy increases

B. Collision frequency increases

C. Enthalpy increases

D. Equilibrium shifts

Q16. Which element shows maximum number of oxidation states? (PYQ NEET 2019)

A. Fe

B. Mn

C. Cu

D. Zn

Q17. Colour of transition metals is due to:

A. d–d transition

B. s–p transition

C. Charge transfer

D. Nuclear transition

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Q18. Lanthanides show contraction due to:

A. Poor shielding of 4f electrons

B. High nuclear charge

C. Increase in atomic size

D. Presence of d electrons

Q19. Which ion is colourless?

A. Cu^{2+}

B. Mn^{2+}

C. Zn^{2+}

D. Fe^{3+}

Q20. Which transition metal is used as catalyst in Haber process? (PYQ NEET 2021)

A. Copper

B. Iron

C. Nickel

D. Platinum

Q21. Most stable oxidation state of lanthanides is:

A. +2

B. +3

C. +4

D. +5

Q22. Coordination number of central metal ion in $[Fe(CN)_6]^{3-}$ is:

A. 4

B. 6

C. 2

D. 3

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Q23. Ligand which donates more than one pair of electrons is called:

A. Monodentate

B. Bidentate

C. Ambidentate

D. Polydentate

Q24. Which of the following is a chelating ligand? (PYQ NEET 2020)

A. NH_3

B. H_2O

C. EDTA

D. Cl^-

Q25. Werner's theory explains:
A. Bonding in organic compounds
B. Isomerism in coordination compounds
C. Metallic bonding
D. Ionic bonding

Q26. Which reaction is used to prepare alkyl halide from alcohol?
A. Dehydration
B. Substitution
C. Elimination
D. Oxidation

Q27. SN1 reaction follows which order? (PYQ NEET 2019)
A. Zero order
B. First order
C. Second order
D. Third order

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Q28. Which alkyl halide reacts fastest in SN1 reaction?
A. CH_3Cl
B. C_2H_5Cl
C. $(CH_3)_3CCl$
D. $CH_3CH_2CH_2Cl$

Q29. Aryl halides are less reactive than alkyl halides due to: (PYQ CUET UG 2023)
A. Partial double bond character
B. Steric hindrance
C. High electronegativity
D. Resonance only

Q30. Which reagent is used in Wurtz reaction? (PYQ NEET 2020)

A. Zn

B. Na

C. Mg

D. Al

Q31. Chlorobenzene does not undergo SN1 reaction because:

A. Strong C–Cl bond

B. Resonance

C. sp^2 hybridisation

D. All of the above

Q32. Functional group of alcohol is: (PYQ CUET UG 2023)

A. –COOH

B. –OH

C. –CHO

D. –CO–

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Q33. Phenol is acidic due to:

A. Inductive effect

B. Resonance stabilization

C. Hyperconjugation

D. Steric effect

Q34. Which compound gives iodoform test? (PYQ NEET 2020)

A. Methanol

B. Ethanol

C. Phenol

D. Ether

Q35. Dehydration of alcohol leads to formation of: (PYQ CUET UG 2022)
A. Alkane
B. Alkene
C. Aldehyde
D. Acid

Q36. Williamson synthesis is used to prepare:
A. Alcohol
B. Ether
C. Aldehyde
D. Ketone

Q37. Phenol reacts with NaOH to form:
A. Sodium phenoxide
B. Sodium ethoxide
C. Sodium benzoate
D. Benzene
b. Maximum in situation (B)
c. Maximum in situation (C)
d. The same in all situations

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Q38. Order of acidity is: (PYQ NEET 2019)
A. Alcohol > Phenol
B. Phenol > Alcohol
C. Ether > Phenol
D. Ether > Alcohol

Q39. Ether has which type of linkage?
A. C–C
B. C–O–C
C. C–N
D. O–O

Q40. Functional group of aldehyde is:

A. –COOH

B. –CO–

C. –CHO

D. –OH

Q41. Tollens' reagent is used to test: (PYQ NEET 2019)

A. Ketones

B. Aldehydes

C. Acids

D. Alcohols

Q42. Which compound is most acidic?

A. Alcohol

B. Phenol

C. Carboxylic acid

D. Aldehyde

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Q43. Oxidation of aldehyde gives:

A. Alcohol

B. Ketone

C. Carboxylic acid

D. Alkene

Q44. Amines are basic in nature due to: (PYQ: CUET UG 2023)

A. Lone pair on nitrogen

B. Presence of H atom

C. Resonance

D. Inductive effect only

Q45. Which amine is most basic in aqueous solution? (PYQ NEET 2020)

A. NH_3

B. Primary amine

C. Secondary amine

D. Tertiary amine

Q46. Glucose is an example of: (PYQ CUET UG 2023)

A. Protein

B. Lipid

C. Carbohydrate

D. Vitamin

Q47. Which bond links amino acids in proteins? (PYQ NEET 2019)

A. Glycosidic

B. Peptide

C. Ester

D. Hydrogen

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Q48. DNA contains which sugar? (PYQ CUET UG 2022)

A. Ribose

B. Glucose

C. Deoxyribose

D. Fructose

Q49. Vitamins required in small quantities are called:

A. Macronutrients

B. Micronutrients

C. Enzymes

D. Hormones

Q50. Which is a reducing sugar? (PYQ NEET 2021)

A. Glucose

B. Sucrose

C. Starch

D. Cellulose

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Answer Key

Question number	Answer Key		Question number	Answer Key
1	B		19	A
2	C		20	B
3	A		21	C
4	D		22	D
5	B		23	A
6	C		24	B
7	A		25	C
8	D		26	A
9	B		27	D
10	C		28	B
11	A		29	C
12	D		30	A
13	B		31	B
14	C		32	C
15	A		33	D
16	B		34	A
17	D		35	B
18	C		36	C

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Answer Key

Question number	Answer Key
37	A
38	D
39	B
40	C
41	A
42	D
43	B
44	C
45	A
46	D
47	B
48	C
49	A
50	D

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